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| **Course** | Digital Signal Processing Lab | **Experiment No.** | 03 |
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| **Language** | Python 3.10 | **Date** | ____ / ____ / 2026 |

# Moving-Average Filter, Sampling and Reconstruction

*Choosing the sampling rate of a bird-song recorder*

***Abstract:*** A real 5-second bird-song field recording (ESC-50 dataset) with a 99 % bandwidth of 6.63 kHz was treated as the analog signal and sampled at 16, 8 and 4 kHz, with no prefilter, a moving-average prefilter and a proper FIR anti-aliasing filter, then reconstructed by zero-order hold, linear and sinc interpolation. Sampling above the Nyquist rate (13.3 kHz) at 16 kHz with an FIR prefilter and sinc reconstruction reached 33.4 dB SNR, while 4 kHz folded 88 % of the song's energy into false tones. The moving average proved to be a cheap but weak anti-aliasing filter: it helped at 4 kHz (-8.6 to -3.4 dB) but hurt at 8 kHz (17.0 to 14.3 dB) because it attenuates the passband. Reconstruction quality ranked sinc > linear > hold by 12 to 27 dB.

***Keywords:*** *sampling theorem, Nyquist rate, aliasing, moving-average filter, anti-aliasing, interpolation, reconstruction, bioacoustics, Python*

## 1. Aim and Objectives

**Aim:** to study the moving-average filter, the sampling of a signal, and the reconstruction of the signal from its samples, using a real audio recording in Python.

- Measure the bandwidth of a real signal and find its Nyquist rate.
- Sample it above and below the Nyquist rate and observe aliasing, with and without an anti-aliasing prefilter (moving average versus FIR).
- Reconstruct the signal by zero-order hold, linear and sinc interpolation and compare their SNR.

## 2. Real-World Application

Ecologists survey birds with small battery-powered acoustic recorders left in a forest for weeks. The sampling rate is the key setting: too high and the battery and memory card run out, too low and the song is destroyed by aliasing. The same trade-off appears in every digital audio system (Table 1). Before sampling, a low-pass **anti-aliasing filter** must remove everything above half the sampling rate; a moving average is the cheapest such filter (only additions), which is why it is common on microcontrollers. After sampling, a **reconstruction** filter turns the samples back into a smooth signal in the playback DAC.

**Table 1. Sampling rates used in practice**

| System | Sampling rate | Usable band | Storage (16-bit, 1 h) |
|---|---|---|---|
| **Narrowband telephone (G.711)** | 8 kHz | 0.3 to 3.4 kHz | 57.6 MB |
| **Wideband / VoIP speech (G.722)** | 16 kHz | 0.05 to 7 kHz | 115.2 MB |
| **Low-power IoT microphone** | 4 kHz | up to 2 kHz | 28.8 MB |
| **Professional audio / video** | 48 kHz | up to 20 kHz | 345.6 MB |
| **Bioacoustic field recorder** | 8 to 384 kHz (selectable) | depends on species | chosen per survey |

## 3. Theory

**Sampling.** Sampling a continuous signal $x_a(t)$ every $T_s = 1/f_s$ seconds gives $x[n] = x_a(nT_s)$. By the Nyquist-Shannon sampling theorem, a signal band-limited to $B$ Hz is recovered exactly if

$$f_s > 2B \qquad (\text{Nyquist rate } = 2B) \tag{1}$$

If $f_s$ is too low, a component at frequency $f$ above $f_s/2$ is not lost but reappears at the **alias** frequency

$$f_{\text{alias}} = |f - kf_s|, \quad k = \text{round}(f/f_s) \tag{2}$$

which is indistinguishable from a genuine tone, so aliasing must be prevented *before* sampling by a low-pass prefilter.

**Moving-average filter.** The $M$-point moving average and its frequency response are

$$y[n] = \frac{1}{M}\sum_{k=0}^{M-1} x[n-k], \qquad |H(e^{j\omega})| = \left|\frac{\sin(M\omega/2)}{M\sin(\omega/2)}\right| \tag{3}$$

with $\omega = 2\pi f/f_s$. It has unity gain at DC and nulls at $f = kf_s/M$, but its passband droops and its first sidelobe is only about 13 dB down, so it is a crude low-pass filter.

**Reconstruction.** The ideal reconstruction is the Whittaker-Shannon interpolation

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n]\,\text{sinc}\left(\frac{t-nT_s}{T_s}\right) \tag{4}$$

Practical systems approximate it by the kernels in Table 2. Quality is scored by SNR $= 10\log_{10}(\sum r^2/\sum(r-\hat{x})^2)$ against the ideal band-limited signal $r$, i.e. the best any system could deliver at that sampling rate.

**Table 2. Reconstruction methods**

| Method | Kernel between samples | Frequency behaviour | Typical use |
|---|---|---|---|
| **Zero-order hold** | hold the last value (staircase) | sinc-shaped droop, strong images | simple DACs |
| **Linear** | straight lines (triangle kernel) | sinc² droop, weaker images | microcontrollers, plots |
| **Sinc (band-limited)** | windowed sinc, ±32 samples | flat to fs/2, images removed | audio interpolators |

> **Note.** An analog recording cannot be stored in a computer, so the 44.1 kHz clip is first resampled to a 48 kHz grid that stands in for the analog signal; every lower rate is then an exact integer decimation (D = 3, 6, 12 for 16, 8, 4 kHz). The bird song has almost no energy near 24 kHz, so the 48 kHz grid is effectively continuous for this study.

## 4. Dataset Description

**ESC-50** (K. J. Piczak, 2015) is a public benchmark of 2,000 five-second environmental recordings in 50 classes, collected from the Freesound field-recording archive and widely used for sound classification research. Clip `1-34495-A-14` (class 14, *chirping birds*) is a single 431 kB WAV file downloaded once from the project's GitHub repository and cached locally. It contains three loud bursts of song between 2.3 and 3.6 s over a quiet background, which makes both the waveform and the spectrum easy to read.

**Table 3. Dataset attributes**

| Property | Value | Property | Value |
|---|---|---|---|
| Source | ESC-50, fold 1 | Clip / class | 1-34495-A-14 / chirping birds |
| Sampling rate | 44,100 Hz | Duration | 5.0 s (220,500 samples) |
| Format | 16-bit PCM, mono | File size | 431 kB |
| Peak / RMS level | 0.594 / 0.085 FS | Spectral peak | 2,102 Hz |
| 99 % bandwidth B | 6,631 Hz | Nyquist rate 2B | 13,262 Hz |
| Energy above 8 / 4 / 2 kHz | 0.7 / 2.0 / 88.2 % | Licence | CC BY-NC (Freesound) |

**Table 4. Raw samples around the loudest point of the recording**

| n (44.1 kHz) | Time (s) | Sample (int16) | Normalised value |
|---|---|---|---|
| 122,404 | 2.775601 | -12823 | -0.3913 |
| 122,405 | 2.775624 | -17080 | -0.5212 |
| 122,406 | 2.775646 | -19318 | -0.5895 |
| 122,407 | 2.775669 | -19478 | -0.5944 |
| 122,408 | 2.775692 | -16943 | -0.5171 |
| 122,409 | 2.775714 | -11959 | -0.3650 |

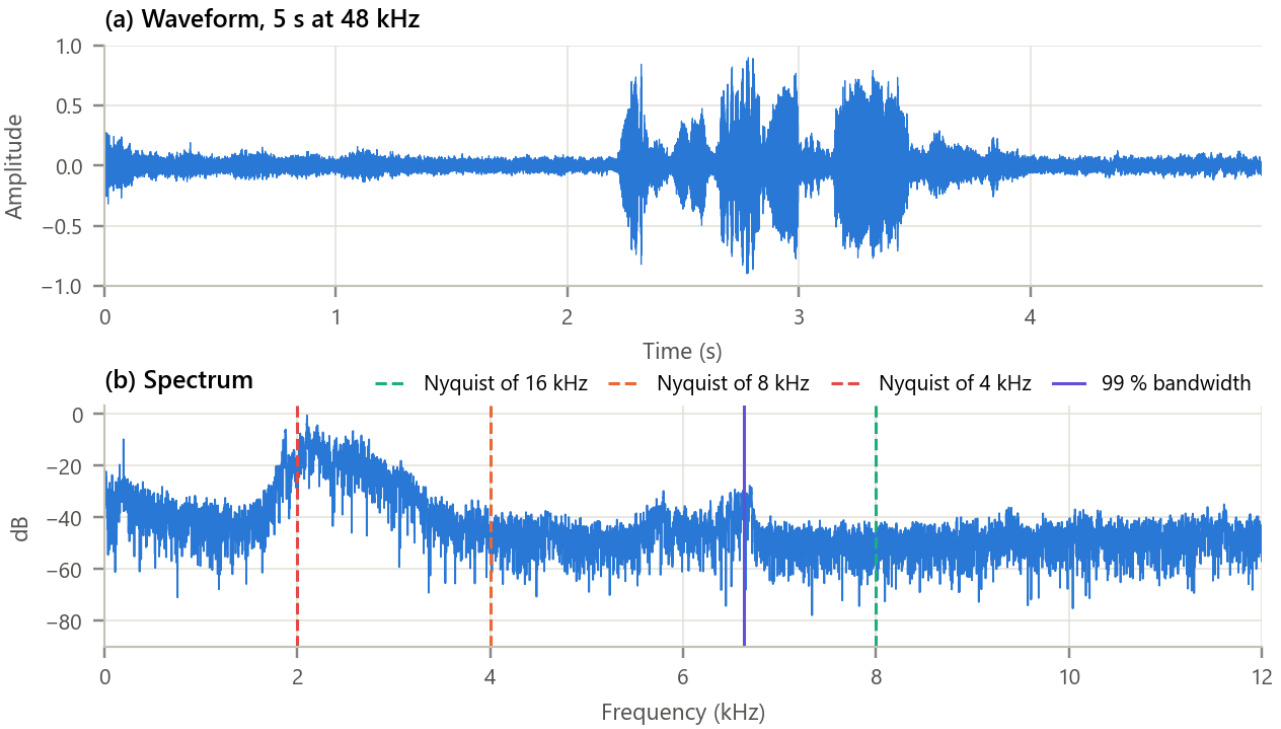



**Figure 1. The recording in time and frequency.** (a) Waveform; (b) spectrum with the Nyquist frequency of each test rate and the 99 % bandwidth.

**Inference.** The song energy sits mainly between 1.8 and 3 kHz with a peak at 2,102 Hz and harmonics reaching about 6.6 kHz. The 16 kHz rate (Nyquist 8 kHz) lies beyond the whole song; 8 kHz (Nyquist 4 kHz) cuts off the upper harmonics (2.0 % of the energy); 4 kHz (Nyquist 2 kHz) lies *below* the main peak and would discard or alias 88 % of the energy. The Nyquist rate of this signal is therefore about 13.3 kHz.

## 5. Methodology and Implementation

For each rate the reference is (1) optionally prefiltered, (2) decimated by D, (3) reconstructed on the 48 kHz grid by each method, and (4) compared with the ideal band-limited signal. The moving-average length is matched to D (M = D), and the FIR filter has 255 taps with its edge at 90% of the new Nyquist frequency.

**Table 5. Software modules**

| Module | Responsibility |
|---|---|
| `config.py` | Clip URL, sampling rates, filter and reconstruction parameters |
| `data_loader.py` | Download once, cache WAV, convert to float, resample to the 48 kHz reference |
| `sampling.py` | Moving average and its response, prefilters, sampling, three reconstructions, SNR |
| `analysis.py` | Bandwidth, prefilter and reconstruction studies, storage cost |
| `plots.py` / `gui.py` | Report figures / interactive sampling lab with audio playback |
| `test_sampling.py` | 10 unit tests (nulls of H, alias frequency, interpolation order), all pass |

**Core of sampling.py**

```
def moving_average(x, M):                          # y[n] = (1/M) sum x[n-k]
    return np.convolve(x, np.full(M, 1.0 / M))[(M - 1) // 2:][:len(x)]

def sample(x, fs_ref, fs_new, kind="None"):        # x_s[n] = x_f(n Ts)
    D = fs_ref // fs_new
    return prefilter(x, fs_ref, fs_new, kind)[::D]

def reconstruct(xs, D, N, method):                 # back to the 48 kHz grid
    if method == "Zero-order hold":
        return np.repeat(xs, D)[:N]
    if method == "Linear":
        return np.interp(np.arange(N) / D, np.arange(len(xs)), xs)
    up = np.zeros(len(xs) * D); up[::D] = xs       # Whittaker-Shannon:
    return oaconvolve(up, sinc_kernel(D), mode="same")[:N]   # sum x[n] sinc((t - nTs)/Ts)

```

**GUI.** `gui.py` (Figure 2) lets the user choose any sampling rate from 2 to 24 kHz, a prefilter (none, moving average of any length, or FIR) and a reconstruction method. It shows 5 ms of the chirp with the samples and the reconstruction, the spectrum with the Nyquist line, the SNR, the aliased energy and the storage cost, and two **Play** buttons let the student *hear* the original and the sampled version. Run `python run_experiment.py` and `python gui.py`.

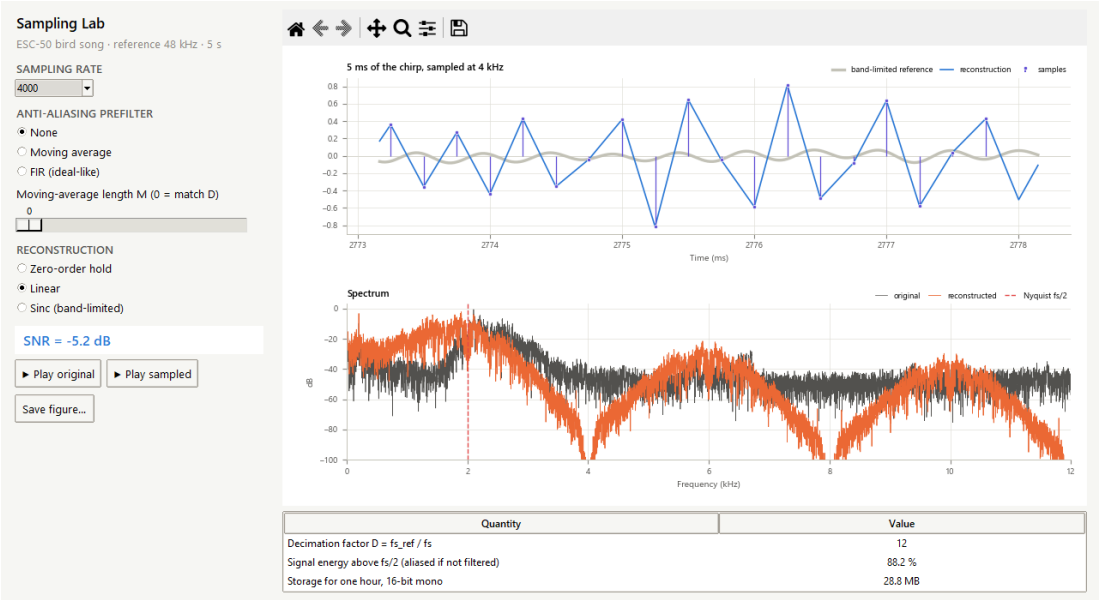



**Figure 2. The sampling lab at 4 kHz with no prefilter and linear reconstruction.** The reconstruction oscillates at the alias frequency while the true band-limited signal is small; the SNR is negative.

## 6. Calculations

Four consecutive samples of the song at 8 kHz (n = 22413 to 22416) are x[n] = {-0.3931, 0.9078, -0.4098, -0.3740}.

**Table 6. Hand calculation on real samples**

| Quantity | Working | Result |
|---|---|---|
| 3-point MA, y[2] | (-0.3931 + 0.9078 + -0.4098) / 3 | 0.0350 |
| 3-point MA, y[3] | (0.9078 + -0.4098 + -0.3740) / 3 | 0.0413 |
| ZOH value at t = (n + ½)Ts | hold x[n] | 0.9078 |
| Linear value at t = (n + ½)Ts | (0.9078 + -0.4098) / 2 | 0.2490 |
| Alias of the song peak at fs = 4 kHz | $\|2102 - 1 \times 4000\|$ | 1898 Hz |
| MA (M = 6) gain at the 8 kHz Nyquist | $\sin(6\omega/2)/(6\sin(\omega/2))$, $\omega = 2\pi \cdot 4000/48000$ | 0.644 (-3.8 dB) |
| Storage at 8 kHz, 1 h | 8000 × 2 bytes × 3600 s | 57.6 MB |

The MA gain of 0.644 at the folding frequency means that components just above 4 kHz are attenuated by less than 4 dB before they alias, which explains the weak anti-aliasing measured below. All values agree with the program output (checked by `test_sampling.py`).

## 7. Results and Discussion

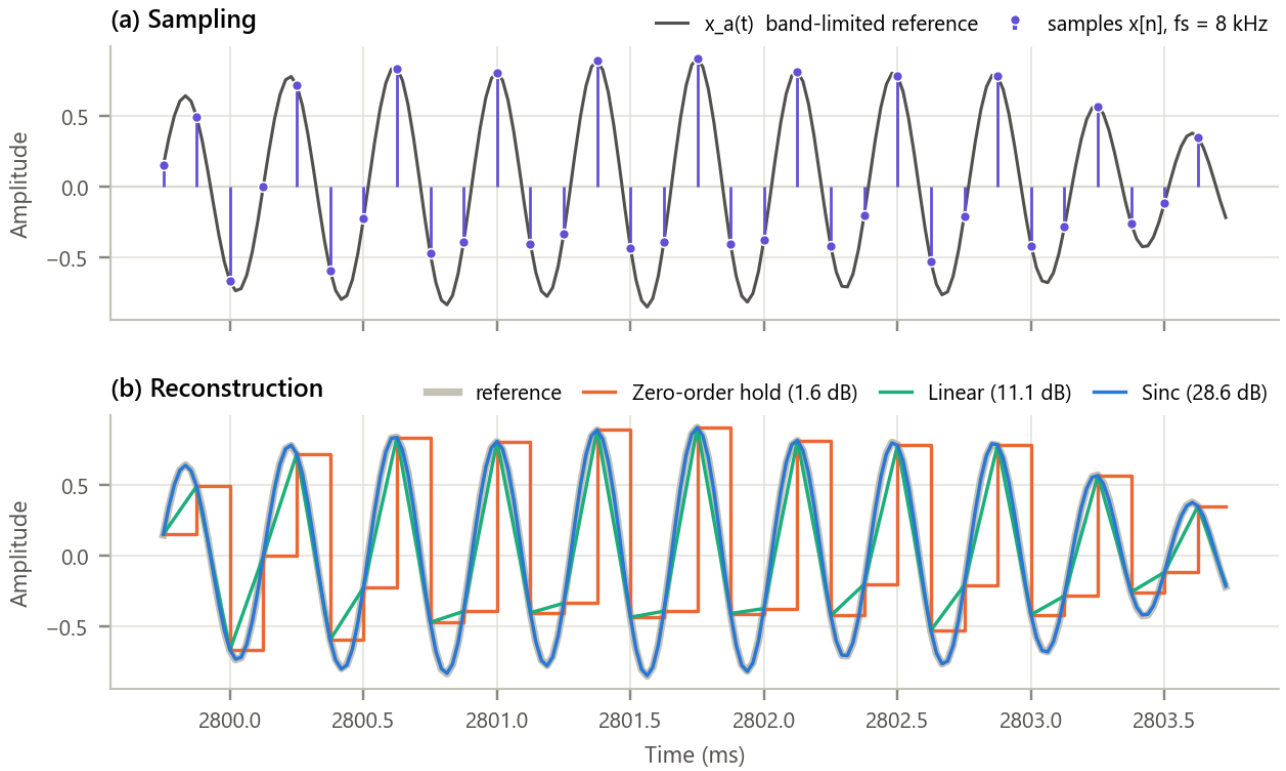



**Figure 3. Sampling at 8 kHz and the three reconstructions (4 ms of one chirp).** (a) Band-limited reference and its samples; (b) zero-order hold, linear and sinc reconstruction with their SNR over the whole clip.

**Inference.**

- At 8 kHz the 2.1 kHz chirp gets only about 3.8 samples per cycle, close to the 2-sample limit, so the way the gaps are filled matters a great deal.
- The **zero-order hold** staircase lags by half a sample and has large steps (1.6 dB); **linear** interpolation cuts the peaks (11.1 dB); the **sinc** interpolator follows the true curve (28.6 dB) because it is the exact inverse of band-limited sampling.

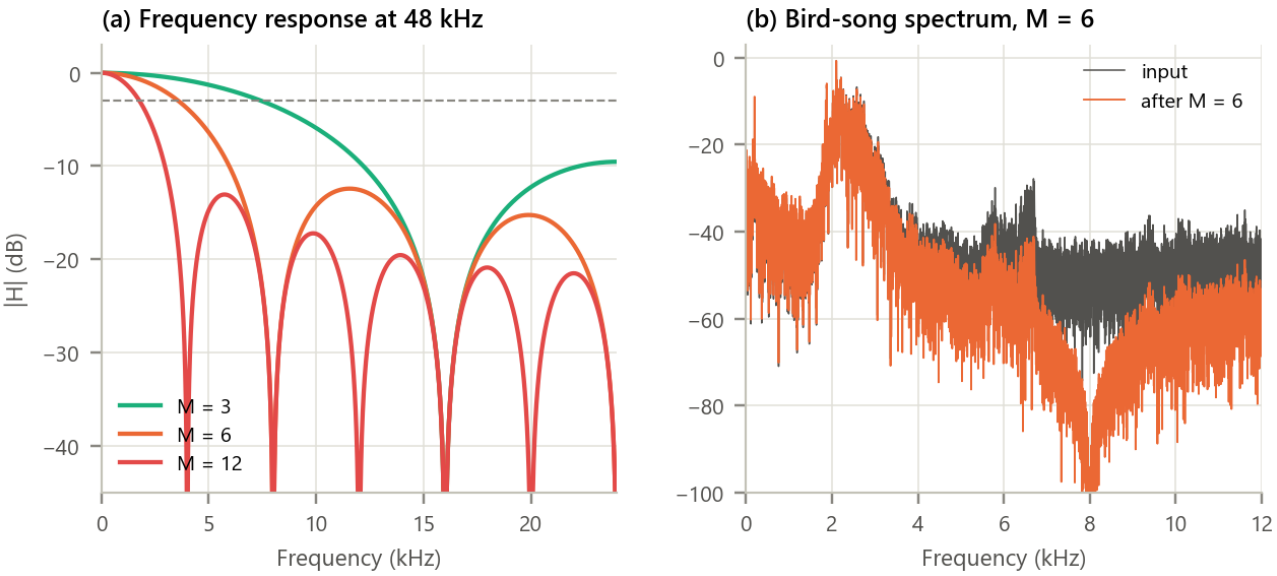



**Figure 4. The moving-average filter.** (a) |H| for M = 3, 6, 12 at 48 kHz; (b) spectrum of the song before and after M = 6.

**Inference.** Each response has nulls at multiples of 48/M kHz and a −3 dB frequency of 7.45, 3.59 and 1.78 kHz for M = 3, 6, 12, i.e. about 0.44·fs/M. The first sidelobe is only 12.4 dB down. On the song, M = 6 places a deep null at 8 kHz and leaves the 2 to 3 kHz song almost untouched, but the 9 to 12 kHz band is attenuated by only 10 to 20 dB because of the sidelobes: the moving average smooths, but it is not a sharp low-pass filter.

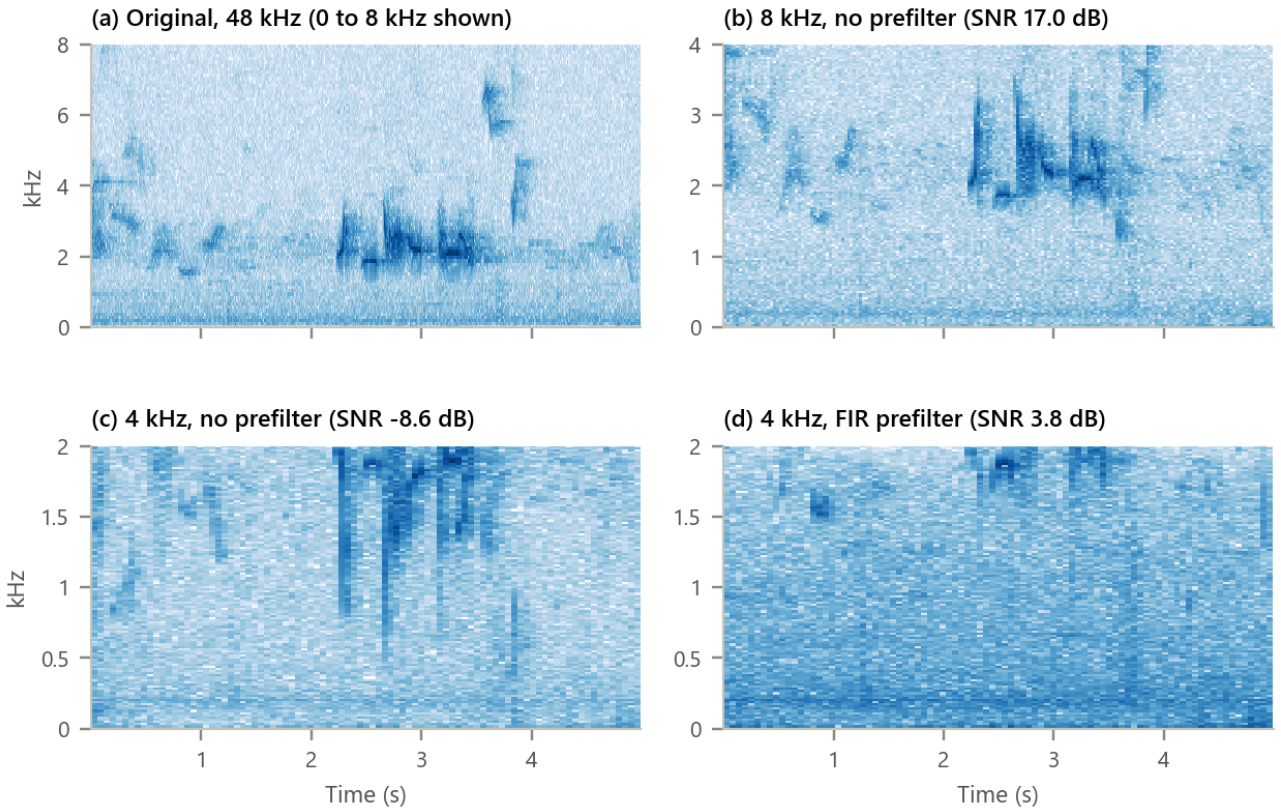



**Figure 5. Aliasing seen in spectrograms.** (a) Original; (b) 8 kHz without prefilter; (c) 4 kHz without prefilter; (d) 4 kHz with the FIR prefilter.

**Inference.** At 8 kHz the song survives with only faint folded harmonics. At 4 kHz without a prefilter the main 2 to 3 kHz song is reflected about 2 kHz: the peak at 2,102 Hz appears at 1898 Hz and strong false chirps fill the 1 to 2 kHz band where the original has none, so an analyst could mis-identify the species. With the FIR prefilter those false chirps vanish, leaving a mostly empty, honest band: the information is lost, but it is not falsified.

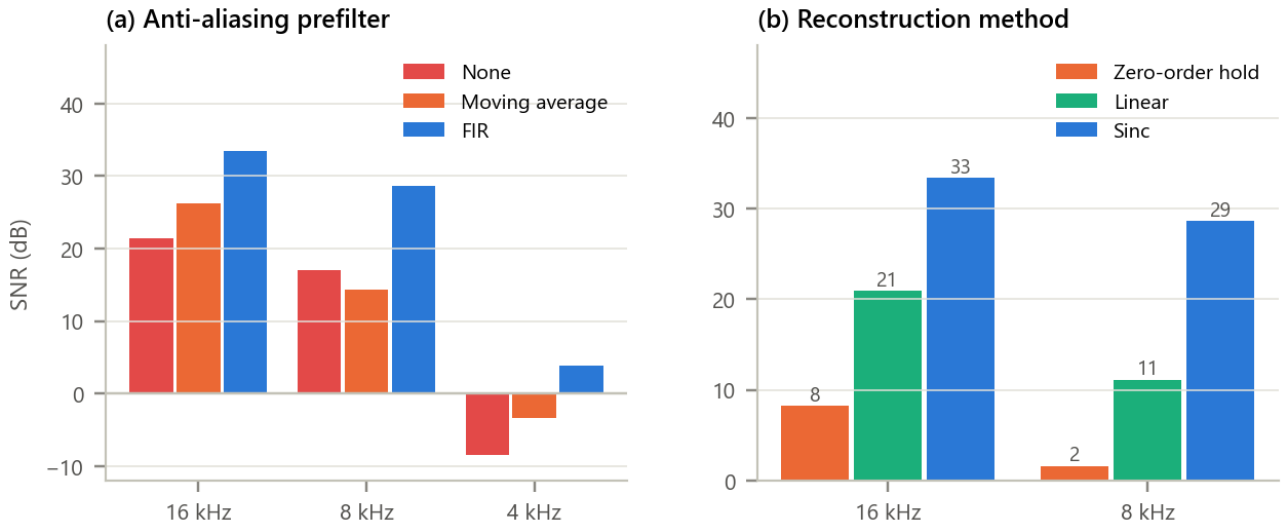



**Figure 6. SNR summary.** (a) Effect of the anti-aliasing prefilter (sinc reconstruction); (b) effect of the reconstruction method (FIR prefilter).

**Inference.** The FIR prefilter is best at every rate (33.4, 28.6 and 3.8 dB). The moving average helps at 16 kHz (+4.8 dB) and 4 kHz, but costs 2.7 dB at 8 kHz, because its passband droop removes more of the song than the little aliasing it prevents. Among reconstructions, sinc beats linear by 12.5 dB and hold by 25.2 dB at 16 kHz.

**Table 7. Summary of results (SNR in dB)**

| Rate | Energy > fs/2 | SNR none | SNR MA | SNR FIR | Best recon. | MB / hour |
|---|---|---|---|---|---|---|
| 16 kHz | 0.7 % | 21.4 | 26.2 | 33.4 | 33.4 dB | 115.2 |
| 8 kHz | 2.0 % | 17.0 | 14.3 | 28.6 | 28.6 dB | 57.6 |
| 4 kHz | 88.2 % | -8.6 | -3.4 | 3.8 | n/a | 28.8 |

## 8. Conclusion

The bird-song recording needs a sampling rate above about 13.3 kHz. At 16 kHz, with a good anti-aliasing filter and sinc reconstruction, it is recovered with 33.4 dB SNR at a cost of 115.2 MB per hour; 8 kHz halves the storage and still keeps the song recognisable; 4 kHz is below the song's main frequency and, without a prefilter, produces convincing but false chirps. The moving-average filter is easy to implement and has exact nulls at multiples of fs/M, but its drooping passband and high sidelobes make it a poor anti-aliasing filter, sometimes worse than none. For a field recorder the right design is therefore: choose fs above 2B for the target species, use a sharp FIR prefilter, and use band-limited (sinc) interpolation on playback.

## 9. Learning Outcomes

- Determine a real signal's bandwidth and Nyquist rate from its spectrum.
- Predict and recognise aliasing, and compute alias frequencies.
- Derive and interpret the frequency response of the moving-average filter and its limits as a low-pass filter.
- Compare zero-order hold, linear and sinc reconstruction quantitatively with SNR.

## 10. References

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[3] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. ACM Multimedia*, 2015, pp. 1015-1018.

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[5] ITU-T Recommendations G.711 (1988) and G.722 (2012), speech coding at 8 and 16 kHz.